

# Worksheet 2A: Lines, Coordinates & Vectors

AIML Foundations Mathematics

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## Part A: Coordinate Basics (10 problems)

1. Plot the following points on a coordinate plane and label each with its coordinates:
  - $A = (3, 4)$
  - $B = (-2, 5)$
  - $C = (-4, -3)$
  - $D = (5, -2)$
  - $E = (0, 3)$
  - $F = (-3, 0)$
2. In which quadrant (I, II, III, IV) or on which axis does each point lie?
  - (a)  $(7, -2)$
  - (b)  $(-3, -8)$
  - (c)  $(0, 5)$
  - (d)  $(-4, 6)$
  - (e)  $(2, 0)$
3. Find the distance between each pair of points:
  - (a)  $(0, 0)$  and  $(3, 4)$
  - (b)  $(1, 2)$  and  $(4, 6)$
  - (c)  $(-2, 3)$  and  $(4, -1)$
  - (d)  $(-5, -2)$  and  $(3, 4)$
4. Find the midpoint of the segment connecting:
  - (a)  $(2, 8)$  and  $(6, 2)$
  - (b)  $(-3, 4)$  and  $(5, -2)$
  - (c)  $(0, 7)$  and  $(-4, -1)$
5. A point starts at  $(2, 3)$  and moves 4 units right and 5 units down. What are its new coordinates?
6. The midpoint of segment PQ is  $(3, 1)$ . If  $P = (7, 4)$ , what are the coordinates of Q?

## Part B: Vectors (12 problems)

Vectors are written in bold:  $\mathbf{v}$ ,  $\mathbf{w}$ ,  $\mathbf{u}$ . They have both magnitude (length) and direction.

7. Given  $\mathbf{v} = (3, 4)$  and  $\mathbf{w} = (1, -2)$ , calculate:
  - (a)  $\mathbf{v} + \mathbf{w}$
  - (b)  $\mathbf{v} - \mathbf{w}$
  - (c)  $\mathbf{w} - \mathbf{v}$
  - (d)  $2\mathbf{v}$
  - (e)  $-3\mathbf{w}$
  - (f)  $2\mathbf{v} + 3\mathbf{w}$
8. Find the magnitude (length) of each vector. Recall:  $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2}$ 
  - (a)  $\mathbf{v} = (3, 4)$
  - (b)  $\mathbf{w} = (5, 12)$
  - (c)  $\mathbf{u} = (-2, 2)$
  - (d)  $\mathbf{p} = (1, 1)$
9. A unit vector has magnitude 1. Find the unit vector in the same direction as:
  - (a)  $\mathbf{v} = (3, 4)$
  - (b)  $\mathbf{w} = (0, 5)$

*(Hint: divide each component by the magnitude)*
10. Vector  $\mathbf{a} = (4, 3)$  and vector  $\mathbf{b} = (-3, 4)$ .
  - (a) Calculate  $\mathbf{a} + \mathbf{b}$
  - (b) Calculate  $|\mathbf{a}|$  and  $|\mathbf{b}|$
  - (c) Calculate  $|\mathbf{a} + \mathbf{b}|$
  - (d) Is  $|\mathbf{a} + \mathbf{b}|$  equal to  $|\mathbf{a}| + |\mathbf{b}|$ ? Why or why not?
11. A drone is at position  $(2, 5)$ . It moves by vector  $\mathbf{v} = (3, -2)$ , then by vector  $\mathbf{w} = (-1, 4)$ . What is its final position?
12. The zero vector is  $\mathbf{0} = (0, 0)$ . For any vector  $\mathbf{v}$ :
  - (a) What is  $\mathbf{v} + \mathbf{0}$ ?
  - (b) What is  $\mathbf{v} + (-\mathbf{v})$ ?
  - (c) What vector  $\mathbf{w}$  satisfies  $\mathbf{v} + \mathbf{w} = \mathbf{0}$ ?
13. Given points  $P = (1, 2)$  and  $Q = (4, 6)$ , find the vector  $\mathbf{PQ}$  that goes from P to Q.

*(Hint:  $\mathbf{PQ} = Q - P$ )*
14. If  $\mathbf{v} = (2, 5)$  and  $\mathbf{w} = (k, 15)$  are parallel (same direction), what is k?

## Part C: Slope and Linear Equations (14 problems)

15. Calculate the slope between each pair of points:
- (a)  $(0, 0)$  and  $(4, 8)$
  - (b)  $(1, 3)$  and  $(5, 11)$
  - (c)  $(2, 7)$  and  $(6, 3)$
  - (d)  $(-3, 2)$  and  $(4, 2)$
  - (e)  $(5, 1)$  and  $(5, 9)$
16. Without calculating, identify the slope as positive, negative, zero, or undefined:
- (a) A line going uphill from left to right
  - (b) A line going downhill from left to right
  - (c) A horizontal line
  - (d) A vertical line
17. Convert to slope-intercept form ( $y = mx + b$ ):
- (a)  $2x + y = 7$
  - (b)  $3x - 2y = 12$
  - (c)  $x + 4y = 8$
  - (d)  $5x - y + 3 = 0$
18. Find the equation of the line with:
- (a) Slope 3 and y-intercept -2
  - (b) Slope  $-\frac{1}{2}$  and y-intercept 4
  - (c) Slope 0 and y-intercept 5
19. Find the equation of the line passing through:
- (a)  $(0, 3)$  with slope 2
  - (b)  $(2, 5)$  with slope -1
  - (c)  $(4, -1)$  with slope  $\frac{3}{4}$
  - (d)  $(-2, 6)$  with slope 0
20. Find the equation of the line passing through both points:
- (a)  $(1, 2)$  and  $(3, 8)$
  - (b)  $(0, 5)$  and  $(4, 1)$
  - (c)  $(-2, -3)$  and  $(4, 9)$
21. Find the x-intercept and y-intercept of each line:
- (a)  $y = 2x - 6$

(b)  $3x + 4y = 12$

(c)  $y = -\frac{1}{2}x + 3$

22. Line A has equation  $y = 3x + 1$ . Line B has equation  $y = 3x - 4$ .

(a) What are their slopes?

(b) Are they parallel, perpendicular, or neither?

(c) Will they ever intersect?

23. Line A has slope 2. What is the slope of a line perpendicular to A?

24. Find the equation of the line:

(a) Parallel to  $y = 2x + 1$  and passing through  $(3, 5)$

(b) Perpendicular to  $y = 3x - 2$  and passing through  $(6, 1)$

## Part D: Systems of Linear Equations (12 problems)

Solve each system by substitution or elimination. State if there's no solution or infinitely many.

25.

$$y = 2x + 1$$

$$y = -x + 7$$

26.

$$x + y = 10$$

$$x - y = 2$$

27.

$$2p + 3q = 13$$

$$p - q = 1$$

28.

$$3m + 2n = 12$$

$$6m + 4n = 24$$

29.

$$2r + 5s = 11$$

$$4r + 10s = 15$$

30.

$$\frac{t}{2} + u = 5$$

$$t - 2u = 4$$

31. Three coffees and two teas cost €11. Two coffees and three teas cost €12. Find the price of each.

32. A rectangle's perimeter is 28 cm. Its length is 4 cm more than its width. Find the dimensions. **Graphical Interpretation:**
33. Without solving algebraically, sketch these lines and estimate their intersection point:

$$y = x + 1$$

$$y = -2x + 7$$

34. What does it mean graphically when a system has:
- (a) Exactly one solution?
  - (b) No solution?
  - (c) Infinitely many solutions?
35. The lines  $y = 2x + 3$  and  $y = 2x - 1$  form a system. Without graphing or substituting, explain why there's no solution.
36. Write a system of two equations that has:
- (a) The solution  $(2, 5)$
  - (b) No solution
  - (c) Infinitely many solutions